Arista Access Node HSR-ACC-69 Routing Protocol Flap Resolution SOP (ALARM_7049)

Node: HSR-ACC-69 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_7049

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a routing protocol flap failure on the Arista Access Node HSR-ACC-69, which is situated at the HSR Layout Access site. The failure domain in question pertains to BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, all of which are critical components of the routing protocol's control plane. The underlying protocol mechanics driving this failure domain are primarily related to BGP's route refresh mechanism, OSPF's neighbor state machine, and ISIS's LSP flooding algorithm. In the context of this access node, these protocols interact with the Arista's Extensible Operating System (EOS) to facilitate routing decisions and packet forwarding. The root cause of this failure domain could be attributed to misconfigured BGP neighbors, OSPF network statements, or ISIS LSP filters, among other possible causes. Therefore, it is essential to meticulously investigate and diagnose the root cause of this failure domain to ensure prompt resolution and minimize network disruption.

Pre-requisite Checks

- Pre-requisite: Verify the site location and node ID to ensure accuracy and dispatch remote hands to HSR Layout Access if necessary.

- Pre-requisite: Check the status of redundant links to ensure proper redundancy and network resilience.

- Pre-requisite: Gather relevant logs and configuration files to facilitate troubleshooting.

Potential Root Causes

Cause	Probability
BGP neighbor misconfiguration	High
OSPF network statement misconfiguration	Medium
ISIS LSP filter misconfiguration	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 12, main routing table version is 12 BGP scan timer is 60 secs Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop In Out LocPrf Weight Packet Route Origin AS_Path *> 192.168.1.0/24 10.10.1.1 0 0 100 100 0 i 65001 192.168.2.0/24 10.10.1.1 0 0 100 100 0 e 65002, Description: This command displays BGP neighbor and routing table information, allowing us to identify any potential misconfigurations.
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```
Command: show ip ospf neighbor, Expected Output: OSPF Process ID: 1 Router ID: 10.10.1.1 Neighbor Count: 2 Neighbor Table:\n 10.10.1.2 FULL/ - 00:00:30 10.10.1.3 FULL/ - 00:00:30, Description: This command displays OSPF neighbor information, enabling us to investigate any issues with OSPF adjacency formation.

Command: show isis database, Expected Output: ISIS Protocol Version: 1.0 ISIS Metric Type: 0x0 ISIS IS-Level: Level-2 ISIS Authentication: None ISIS Adjacency Table:\n 10.10.1.2 20ms FULL/ 00:00:30 10.10.1.3 20ms FULL/ 00:00:30, Description: This command displays ISIS LSP database and adjacency information, allowing us to diagnose any issues with ISIS routing.
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Resolution Steps

Step 1: Step: Verify BGP neighbor configuration and correct any misconfigurations., Command: show ip bgp neighbors, Description: This command displays BGP neighbor information, enabling us to verify the configuration and make any necessary corrections.

Step 2: Step: Verify OSPF network statement configuration and correct any misconfigurations., Command: show ip ospf network, Description: This command displays OSPF network statement information, allowing us to verify the configuration and make any necessary corrections.

Step 3: Step: Verify ISIS LSP filter configuration and correct any misconfigurations., Command: show isis database, Description: This command displays ISIS LSP database information, enabling us to verify the configuration and make any necessary corrections.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 13, main routing table version is 13 BGP scan timer is 60 secs Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop In Out LocPrf Weight Packet Route Origin AS_Path *> 192.168.1.0/24 10.10.1.1 0 0 100 100 0 i 65001 192.168.2.0/24 10.10.1.1 0 0 100 100 0 e 65002, Description: This command verifies that the BGP routing table has been updated and that the root cause has been resolved.

Command: show ip ospf neighbor, Expected Output: OSPF Process ID: 1 Router ID: 10.10.1.1 Neighbor Count: 2 Neighbor Table:\n 10.10.1.2 FULL/ - 00:00:30 10.10.1.3 FULL/ - 00:00:30, Description: This command verifies that OSPF neighbors have formed properly and that the root cause has been resolved.

Command: show isis database, Expected Output: ISIS Protocol Version: 1.0 ISIS Metric Type: 0x0 ISIS IS-Level: Level-2 ISIS Authentication: None ISIS Adjacency Table:\n 10.10.1.2 20ms FULL/ 00:00:30 10.10.1.3 20ms FULL/ 00:00:30, Description: This command verifies that ISIS LSP database has been updated and that the root cause has been resolved.
```

Roll-back Procedures

- Step: Restore the original BGP neighbor configuration., Command: copy running-config startup-config, Description: This command restores the original BGP neighbor configuration, effectively rolling back the changes made during the troubleshooting process.

- Step: Restore the original OSPF network statement configuration., Command: copy running-config startup-config, Description: This command restores the original OSPF network statement configuration, effectively rolling back the changes made during the troubleshooting process.

- Step: Restore the original ISIS LSP filter configuration., Command: copy running-config startup-config, Description: This command restores the original ISIS LSP filter configuration, effectively rolling back the

changes made during the troubleshooting process.